WHAT IS PANDAS?

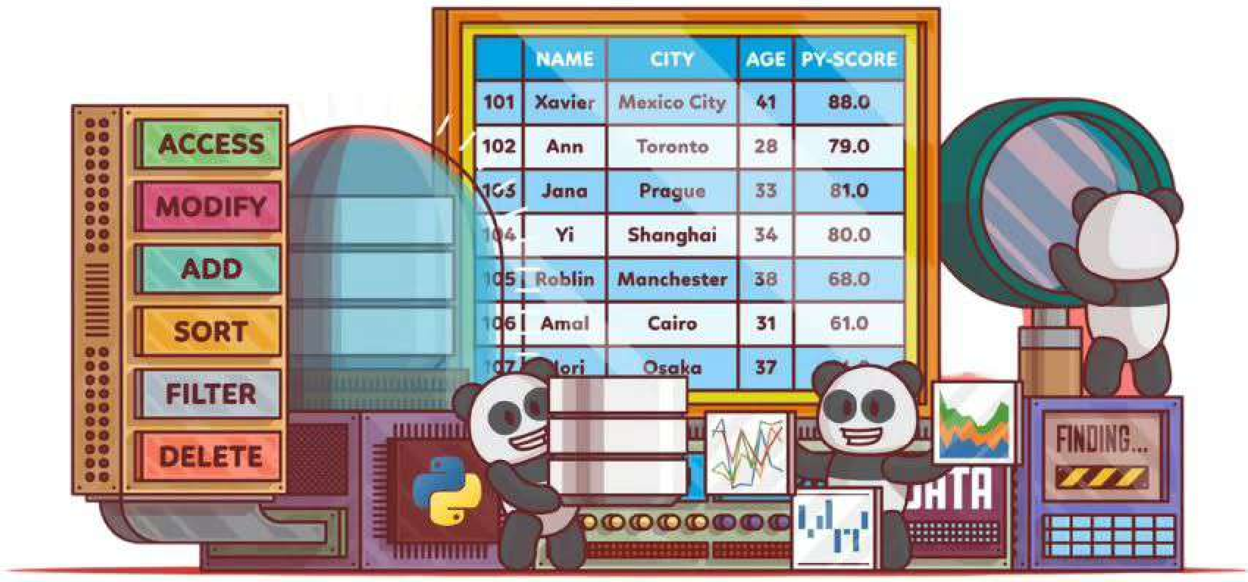
pandas

Explained in fun and easy way

Big Data Specialist

SWIPE

Pandas is a powerful and popular **open-source Python library** designed for data **manipulation** and **analysis**.



It provides data structures and functions that make it easy to work with **structured data**, such as spreadsheets or SQL tables, efficiently and effectively.

The central data structure in Pandas is the **"DataFrame."** Think of it as a two-dimensional table, much like an Excel spreadsheet, where data is organized in rows and columns.

```python
import pandas as pd
movies = pd.read_csv("imdb_movies.csv")
movies.head()
```

id	title	revenue	rating
1	Jurrasic world	$1.672 billion	6.9
2	Mad Max: Fury Road	$378 million	8.1
3	Interstellar	$677.4 million	8.7
4	Oppenheimer	TBD	8.8

DataFrames allow you to store, filter, and manipulate data in a structured manner, making it a go-to tool for data processing tasks.

The library also offers powerful **indexing** and **slicing** capabilities, allowing users to access and modify specific subsets of data efficiently.

```
movies[1:3][["title","rating"]]
```

rows

columns

0 title rating

	title	rating
1	Mad Max: Fury Road	8.1
2	Interstellar	8.7

3

This enables easy data **exploration** and **extraction** of valuable **insights** from large datasets.

Pandas is also equipped with various methods and functions for performing data **operations**.

```
Name      Subset            Value
A         67-A-5678         14
A         58-ABC-87555      187
A         45-ASH-87954      5465
S         34-A-8785         454
S         58-ASO-98978      54
S         23-ASH-87895      784
X         98-X-87876        455
X         87-ABC-54578      4545
X         56-ASH-89667      854
Y         09-D-98644        45
Y         87-ABC-78834      98
Y         87-ASH-87455A     4566
L         67-A-87545        78
L         89-GHS-08753      12
L         78-PHU-09876      655
```

to keep only those groups of rows whose "subset" columns are of pattern; *, *ABC, *ASH (Note: * is any alphabet or digit).

```
>>> filtered
   Name         Subset  Value
0     A      67-A-5678     14
1     A   58-ABC-87555    187
2     A   45-ASH-87954   5465
6     X     98-X-87876    455
7     X   87-ABC-54578   4545
8     X   56-ASH-89667    854
9     Y     09-D-98644     45
10    Y   87-ABC-78834     98
11    Y  87-ASH-87455A   4566
```

```
filtered = df[df.groupby('Name')
['Subset'].transform(lambda x: len(x) >= 3
and'-ABC-' in x.iloc[1]
and '-ASH-' in x.iloc[2])]
```

It can handle data **cleaning** by removing or imputing missing values, data filtering using conditional expressions, and data aggregation with grouping and summarizing capabilities.

Pandas also integrates well with other Python libraries, such as NumPy and Matplotlib, enhancing its capabilities further

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

It can easily **convert** data between different **formats**, including CSV, Excel, SQL databases, and JSON, providing seamless interoperability with external data sources.

pandas ECOSYSTEM...

Statistics and Machine Learning

- pandas-tfrecords
- Feature Tools
- Statsmodel
- Compose
- sklearn-pandas
- Stumpy

APIs

- Pandas-datareader
- Quandl/Python
- PandaSMDX
- Fredapi
- Dataframe_sql

Domain Specific

- Geopandas
- gurobipy-pandas
- staircase
- xarray

Visualization

- Altair
- Plotnine
- Bokeh
- Plotly
- Seaborn
- LUX

IDEs

- IPython
- Jupyter Lab
- Modin-spreadsheet
- Spyder

Out-of-Core

- Blaze
- Clyon
- Dask
- Ibis
- Koalas
- Vaex

and more...

Thanks to its highly optimized and fast implementation, Pandas can **efficiently handle large datasets** without compromising performance.

pandas

This makes it a favorite tool for data scientists, analysts, and engineers when dealing with real-world data challenges.

In summary, Pandas is like a versatile **data powerhouse** that allows Python programmers to manage and analyze data effortlessly, unlocking the full potential of structured data for a wide range of data-driven applications.